Dear Editors:

We would like to submit the enclosed manuscript entitled “Eroding the First Rung: Generative Artificial Intelligence, Organizational Learning, and the Self-Reinforcing Decline of Youth Employment in Chinese Listed Firms” for consideration in the Special Issue “Systems Approaches to Generative AI: Workforce Development, Organisational Learning, and Economic Transformation” of Systems. The manuscript has not been published elsewhere and is not under consideration by any other journal. No conflict of interest exists in its submission, and it has been approved for publication by all authors.

Entry-level work is the first rung of the ladder by which economies manufacture experienced workers, and generative artificial intelligence is most capable at exactly the tasks that rung is made of. The urgency is not hypothetical: China produced a record 12.22 million college graduates in 2025 while youth unemployment stayed above 17 percent, and the international evidence already shows employment of workers in their early twenties falling relative to experienced workers in the same occupations. What no firm-level study has yet asked is whether this is a one-way shock or a loop. Using 25,701 firm-year observations for 3,000 Chinese A-share listed firms over 2016–2024, we show that it is a loop, and we measure how much the loop adds.

The highlights of the paper are as follows.

(1) A displacement that is specific to the first rung. A one-standard-deviation increase in generative-AI application intensity lowers the youth employment share by 1.74 percentage points, 6.0 percent of its mean. The same regression run on employees aged 31 to 50 gives a coefficient 11 times smaller. This is not a contraction of employment; it is a change in its age composition.

(2) The firm stops producing skill, not merely stops hiring. Two channels are measured and both are significant: a contracting routine task base accounts for 22.3 percent of the total effect, and suppressed investment in employee training for a further 12.5 percent, with firm-block bootstrap intervals excluding zero. The second channel does not exist in the task-based framework that dominates this literature, because that framework treats labor as an input to be allocated rather than an asset the firm produces.

(3) The loop closes, and it amplifies. Estimating the adoption–training–youth system as a bias-corrected panel vector autoregression, the return arcs are negative and significant: a firm whose training investment and youth cohort have contracted adopts more generative AI the following year (−0.347 and −0.032, both $p < 0.01$). Switching the return arcs off and re-simulating shows that the loop adds 19.0 percent to the eight-year cumulative displacement. The largest eigenvalue is 0.722, so the trap is bounded rather than explosive — which is precisely what makes it addressable.

(4) A leverage point management can build. The same adoption intensity displaces about half as much youth employment in firms one standard deviation above the mean of organizational learning capability, and correspondingly more where relative labor costs are high. Industry classification does not discriminate between firms; the direct measure of absorptive capacity does. The outcome is therefore organizational rather than technological, and it is manageable.

We believe the manuscript speaks directly to the aims of the Special Issue, which asks for systems approaches to generative AI, workforce development and organizational learning. Our contribution is a systems result in the strict sense. The relationship between generative AI and youth employment is not a treatment effect but a reinforcing feedback structure — Repenning and Sterman’s capability trap, applied for the first time to a firm’s skill-formation system and estimated rather than asserted. The practical consequence is that any single-equation estimate of the effect understates it by about a fifth, and that policy aimed at the moment of adoption operates on the forward arc and is partly undone by the return arc, whereas instruments that protect training provision act on the arc that closes the loop. The design is conventional where it should be — two-way fixed effects, overidentified instruments passing the Kleibergen–Paap and Hansen tests, propensity score matching, entropy balancing, an age placebo, Oster bounds and a wild cluster bootstrap — and novel only where the systems claim requires it. We deeply appreciate your consideration of our manuscript and look forward to receiving comments from the reviewers. Correspondence should be directed to the address below.

Thanks very much for your attention to our paper.

Name: Tiantian Mo
Institution and address: College of Business, Jiaxing University; Jiaxing 314001, China
Email: 00008227@zjxu.edu.cn

Very sincerely yours,
Xinyu Cai and Tiantian Mo